

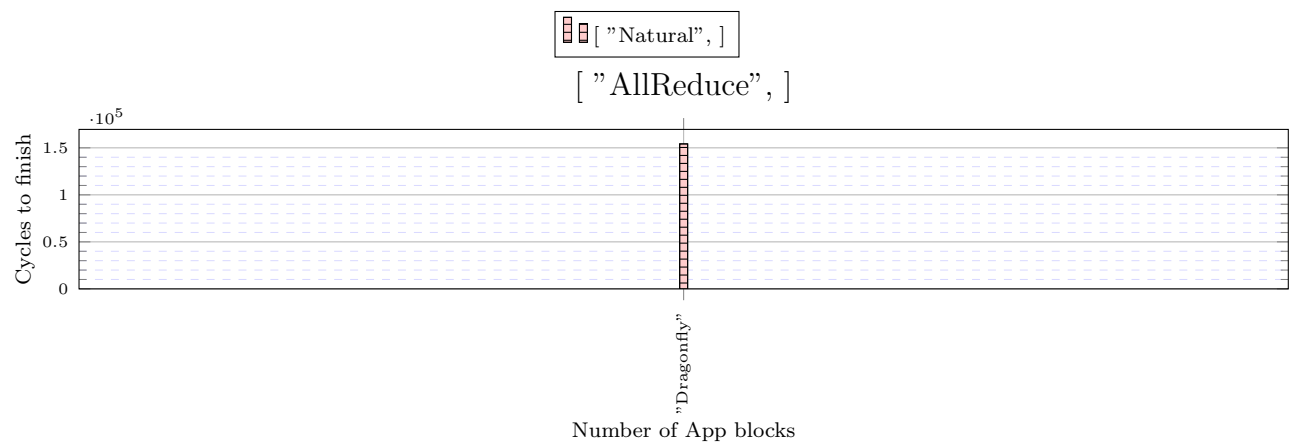


Figure 1: X: ["AllReduce",]

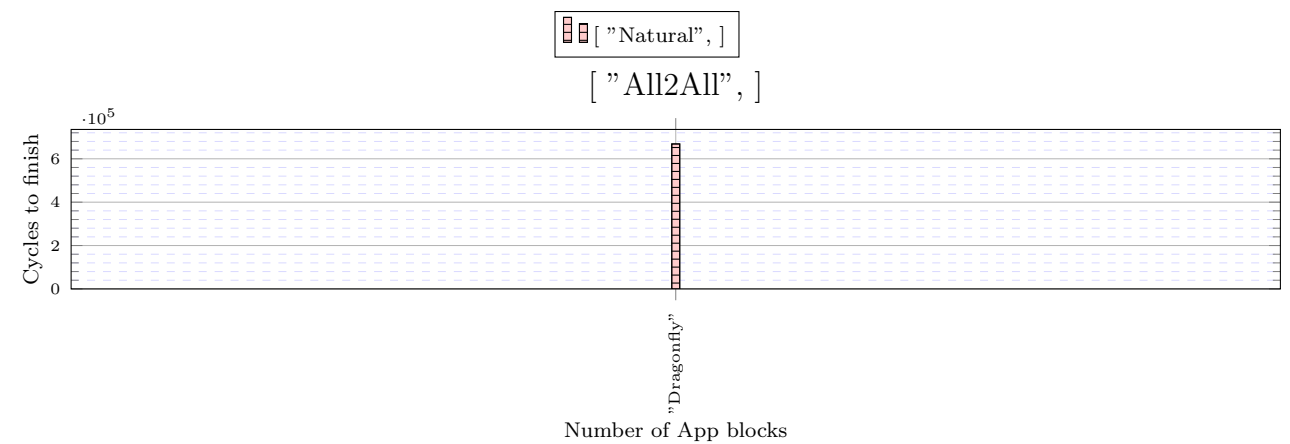


Figure 2: X: ["All2All",]

The following versions used in the simulations.

- heads/master-dirty-ce88a576295ce68f40736599a9e3e04f24f38801(0.6.3)
- heads/master-dirty-65e4f7db651d154c0abb3e3d0a691be7d6d92d82(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 10000000000000000,
  general_frequency_divisor: 2,
  topology: topologia![
    Dragonfly{ global_ports_per_router: 8, servers_per_router: 8, global_arrangement: Palmtree, group_size: 16, number_of_groups: 129, legend_name: "Dragonfly" },
  ],
  traffic: ![
    TrafficMap{
      tasks: 16512,
      map: ![
        Composition{
          patterns: [
            Identity{ legend_name: "Natural" },
            CartesianEmbedding{
              source_sides: [ 16384 ],
              destination_sides: [ 16512 ]},
            middle_sizes: [ 16384, 16512 ],
            legend_name: "Natural" },
          application: ![
            AllReduce{ tasks: 16384, data_size: 16384, legend_name: "AllReduce" },
            legend_name: "AllReduce" },
        ],
      TrafficMap{
        tasks: 16512,
        map: ![
          Identity{ legend_name: "Natural" },
          application: ![
            All2All{ tasks: 16512, data_size: 132096 },
            legend_name: "All2All" },
        ],
      },
    maximum_packet_size: 16,
    router: InputOutput{
      virtual_channels: topologia![ 4 ],
      virtual_channel_policies: topologia![
        [
          MapLabel{
            label_to_policy: [
              Chain{
                policies: [
                  OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true, use_neighbour_space: true, aggregate: false }]],
              Chain{
                policies: [
                  AverageOccupancyFunction{
                    label_coefficient: 0,
                    occupancy_coefficient: 2,
                    product_coefficient: 0,
                    constant_coefficient: 16,
                    use_internal_space: true,
                    use_neighbour_space: true,
                    exclude_minimal_ports: true,
                    virtual_channels: [ 0 ]}],
                  LowestLabel,
                  EnforceFlowControl,
                  Random]],
            ],
          allocator: Random,
          buffer_size: 128,
          bubble: false,
          flit_size: 16,
          intransit_priority: false,
          allow_request_busy_port: true,
          output_buffer_size: 64,
          crossbar_frequency_divisor: 1,
          crossbar_delay: 2},
        ],
      routing: topologia![
        Sum{
          policy: TryBoth,
          first_routing: ChannelMap{
            routing: AscendantChannelsWithLinkClass{
              routing: WeighedShortest{
                class_weight: [ 1, 100 ]},
              bases: [ 1, 1 ]},
            map: [
              [ 0 ],
              [ 1 ]],
            ],
          second_routing: Valiant4Dragonfly{
            first: ChannelMap{
              routing: AscendantChannelsWithLinkClass{
                routing: WeighedShortest{
                  class_weight: [ 1, 100 ]},
                bases: [ 1, 1 ]},
              map: [
                [ 0 ],
                [ 1 ]],
              ],
            second: ChannelMap{
              routing: AscendantChannelsWithLinkClass{
                routing: WeighedShortest{
                  class_weight: [ 1, 100 ]},
                bases: [ 1, 1 ]},
              map: [
                [ 2 ],
                [ 3 ]],
              ],
            ],
          first_allowed_virtual_channels: [ 0, 1, 2, 3 ],
          second_allowed_virtual_channels: [ 0, 1, 2, 3 ],
          second_extra_label: 1,
          legend_name: "Ugal" },
        ],
      link_classes: [
        LinkClass{ delay: 2 },
      ],
    },
  ],
}
```

```
LinkClass{ delay: 2 },
LinkClass{ delay: 2 }],
launch_configurations: [
  Slurm{ job_pack_size: 1, time: "3-00:00:00" }]]
```